Random variable

It is a variable that has probabilities associated with the possible values.

Example 1:

Let a coin be tossed once. $S = \{H, T\}$. Let $X =$ number of heads obtained. Then, X can take values '0' or '1'.

$P(X = 0) = P(T) = 0.5$

$P(X = 1) = P(H) = 0.5$

So, X is a random variable.

Example 2:

Height is a variable. When we are interested in the probabilities of the possible values of height, we call it a random variable.

- Any numerical variable is a random variable when we are interested in the probabilities of its possible values. (Some authors say that categorical variables are also random variables if probability is involved.)

Discrete random variable

A random variable is called discrete if there are gaps between any two possible values. (Therefore, number of possible values is finite or countably infinite.)

Example 3:

Let two fair coins be tossed. $S = \{HH, HT, TH, TT\}$. Let $X =$ number of heads obtained. Then, X is a discrete random variable that can take values $0, 1$ or 2.

$P(X = 0) = P(TT) = 0.25$

$P(X = 1) = P(HT \text{ or } TH) = P(HT) + P(TH) = 0.25 + 0.25 = 0.50$

$P(X = 2) = P(HH) = 0.25$

Here, number of possible values of X is 3 (finite).

Example 4:

Let $X =$ number of calls that comes to a mobile in a day. Then $X = 0, 1, 2, \cdots$. Here, number of possible values of X is <u>countably infinite</u>. (Probabilities of this type of variables will be discussed later.)

Continuous random variable

A random variable for which all values within a certain interval are possible is called a continuous random variable. Such a variable has uncountably infinite number of possible values.

Example 5:

Let $X =$ weight in gram of a football (regulation size). Then, $410 < X < 450$. Here, number of possible values of X is <u>uncountably infinite</u>.

Example 6:

Let $X =$ Lifetime of an electric component. Then, $0 < X < \infty$. Here, number of possible values of X is <u>uncountably infinite</u>.

(Probabilities of continuous variables will be discussed later.)

Probability mass function for discrete random variable

Let X be a discrete random variable. The probability mass function (pmf) or probability function of X, denoted by $p(x)$, is defined as $p(x) = P(X = x)$. It satisfies the following two consitions:

(i) $p(x) \geq 0$

(ii) $\sum_{x} p(x) = 1$

- The function $p(x)$ is also called the probability distribution of X.

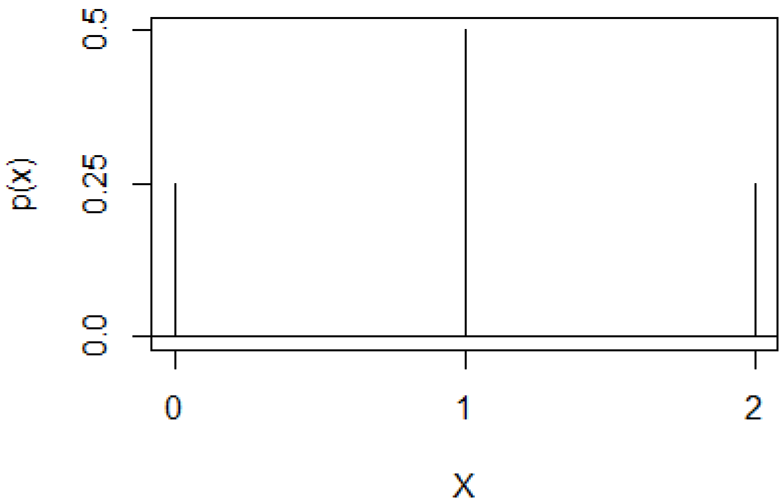
Example 7:

Let a random experiment consist of tossing two coins once. $S = \{HH, HT, TH, TT\}$. Let $X =$ number of heads obtained. Then, X is a discrete random variable that can take values $0, 1$ or 2. The pmf of X is as follows:

x	0	1	2
$p(x)$	0.25	0.50	0.25

Here, $p(0) = P(X = 0) = 0.25$. That is, for input '0', output of the function is 0.25. Similarly, $p(1) = P(X = 1) = 0.50$, and so on.

The figure below shows the function $p(x)$ plotted against x:

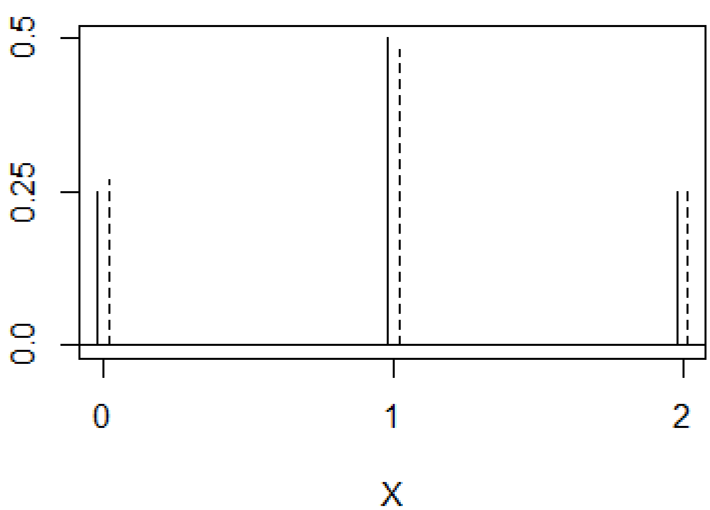


Frequency distribution vs probability distribution

Let the random experiment in Example 7 be repeated $n = 1000$ times, and the data be used to construct the following frequency table.

x	Frequency	Relative Frequency	Probability
0	270	0.27	0.25
1	480	0.48	0.50
2	250	0.25	0.25
Total	1000	1.00	1.00

The last column of the table above uses the probabilities obtained in Example 7. Note that relative frequencies (sample) are comparable to probabilities (population). In the plot below, solid lines are used to show probabilities, while dotted lines are used to show relative frequencies.



Exercise:

Consider the following pmf.

x	1	3	4	9
$p(x)$	0.2	0.5	k	0.1

Determine (a) the value of k (b) $P(X = 3)$ (c) $P(X = 3 \text{ or } 4)$ (d) $P(1.7 < X < 4.1)$ (e) $P(X > 4)$ (f) $P(X \geq 4)$ and (g) $P(X < 1)$.

Solution:

(a) $0.2 + 0.5 + k + 0.1 = 1$

$\therefore k = 0.2$

(b) $P(X = 3) = 0.5$

(c) $P(X = 3 \text{ or } 4) = p(3) + p(4) = 0.7$

(d) $P(0.7 < X < 4.1) = p(1) + p(3) + p(4) = 0.9$

(e) $P(X > 4) = p(9) = 0.1$

(f) $P(X \geq 4) = p(4) + p(9) = 0.3$

(g) $P(X < 1) = 0$.

Probability density function for continuous random variables

Let X be a continuous random variable. The probability density function (pdf) of X, denoted by $f(x)$, is a function that satisfies the following two conditions:

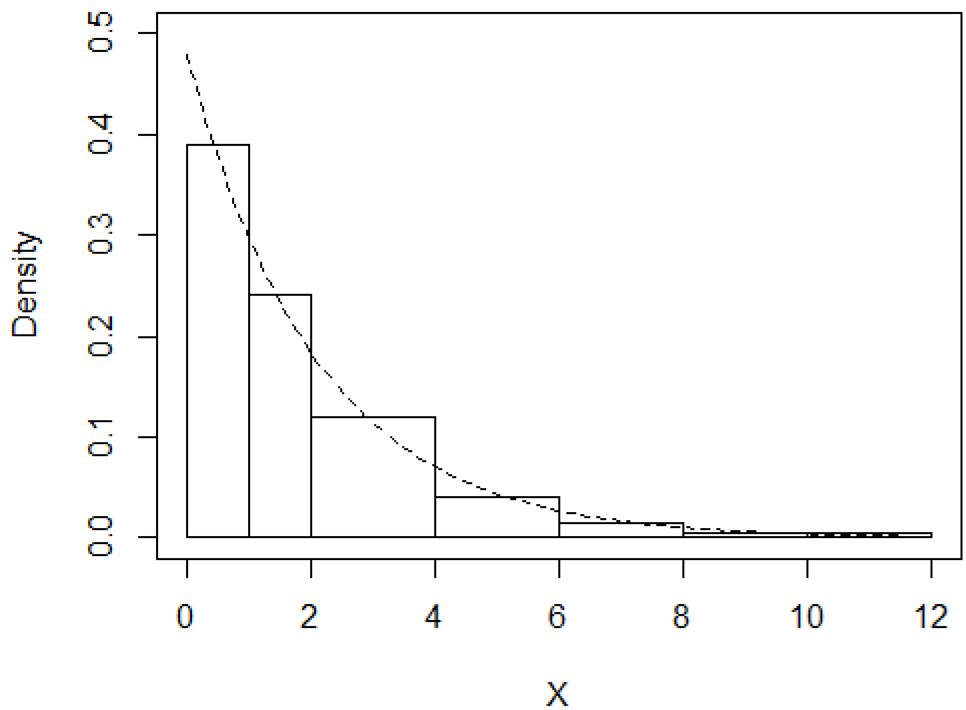
(i) $f(x) \geq 0$.

(ii) $$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

- The function $f(x)$ is also called the probability distribution of X. The term 'probability distribution' is used for both discrete and continuous random variables.
- $f(x)$ represents population while histogram represents sample.

Frequency distribution vs probability distribution

A pdf representing population and a histogram representing sample are shown together in the following plot. We can say that the pdf fits the histogram well. and could be the population from which the sample came.



Example:

Consider the function

$$f(x) = \begin{cases} 0.48\, e^{-0.48\, x}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

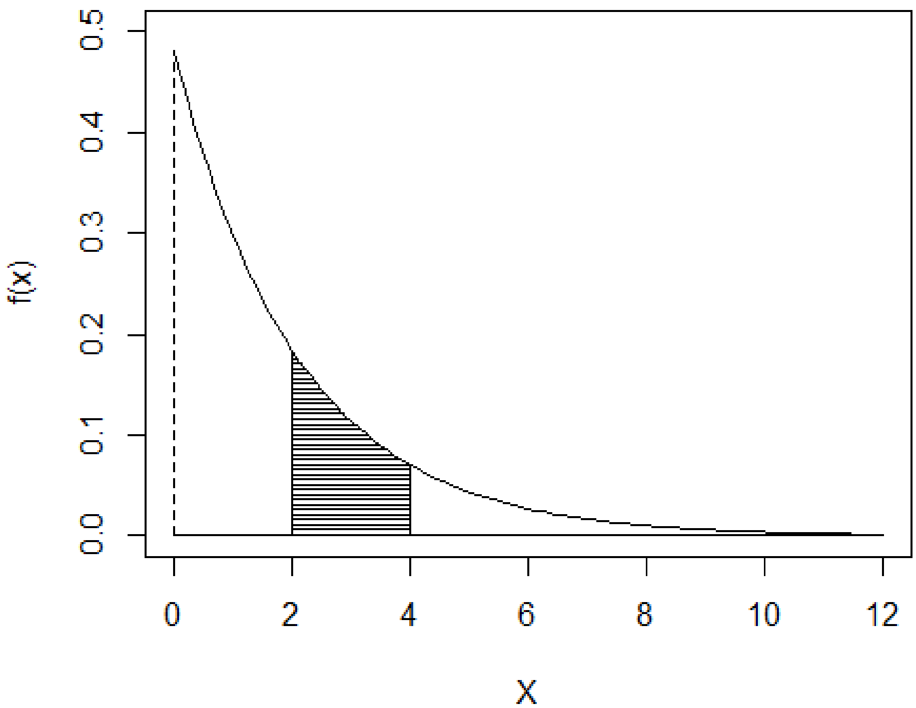
This function is not negative for any value of x. Also,

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-\infty}^{0} 0\, dx + \int_{0}^{\infty} 0.48\, e^{-0.48\, x}\, dx = 1$$

Thus, the function above satisfies both the condition and, therefore, is a pdf.

Calculation of probability from pdf

In a histogram, area of a bar gives the relative frequency of the corresponding class-interval. Similarly, in a pdf, area over an interval gives the probability of that particular interval. In the plot below, the shaded area is equal to $P(2 \le X \le 4)$. That is,



$$P(2 \le X \le 4) = \int_2^4 f(x)\,dx = \int_2^4 0.48\,e^{-0.48\,x}\,dx = 0.236$$

In general, we can calculate probabilities by using the following formula:

$$P(a \le X \le b) = \int_a^b f(x)\,dx$$

Exercise:

Check whether the following function is a pdf.

$f(x) = 2x, \;\; 0 < x < 1.$

Solution:

i)

$f(x) \ge 0 \;$ for $\; 0 < x < 1.$

ii)

$$\int_0^1 f(x)\,dx$$

$$= \int_0^1 2x\,dx$$

$$= \left.\frac{2x^2}{2}\right|_0^1$$

$$= 1$$

Therefore, $f(x)$ is a pdf.

Exercise:

Check whether the following function is a pdf.

$f(x) = x - 0.5, \;\; 0 < x < 2.$

Solution:

Do it yourself.

Exercise:

Consider the following pdf.

$f(x) = k(1 - x^2), \;\; -1 < x < 1.$

Determine (a) the value of k (b) $P(X > 0)$ (c) $P(-0.7 < X < 0.7)$ (d) density at $X = 0.5$ (e) $P(X = 0.5)$ (f) $P(X = 0.8)$.

Solution:

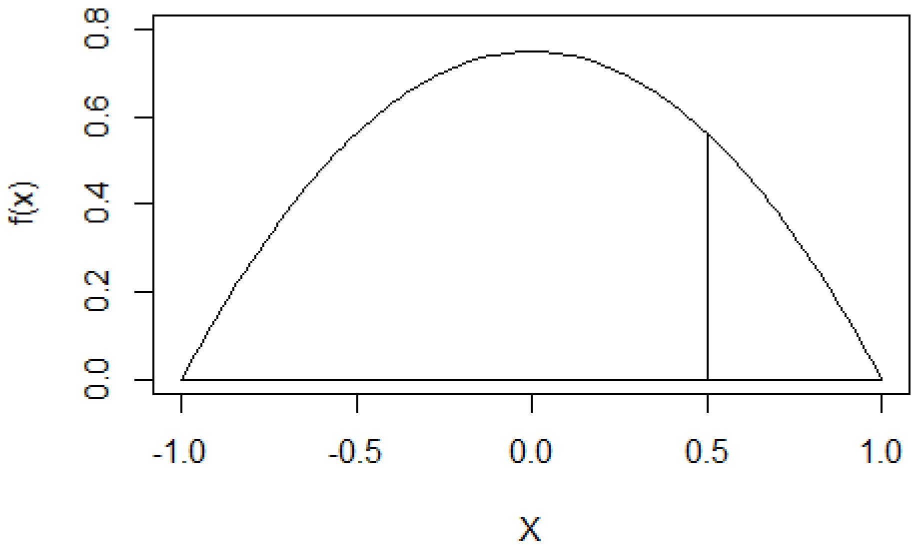
(a)

$$\int_{-1}^1 f(x)\,dx = 1$$

$$\therefore \int_{-1}^1 k(1 - x^2)\,dx = 1$$

$$\therefore k\left(x-\frac{x^3}{3}\right)\Bigg|_{-1}^{1}=1$$

$$\therefore k=\frac{3}{4}=0.75$$



(b)

$P(X>0)$

$$=\int_0^1 0.75(1-x^2)\,dx$$

$=0.5$

(c)

$P(-0.7<X<0.7)$

$$=\int_{-0.7}^{0.7} 0.75(1-x^2)\,dx$$

$=0.8785$

(d)

$f(0.5)=0.75(1-0.5^2)=0.5625$

Therefore, density at $X=0.5$ is 0.5625.

Note:

$f(0.5)$ is the height at $X = 0.5$. Height does not represent probability.

Area (integration) gives probability.

(e)

$P(X = 0.5)$

$$= \int_{0.5}^{0.5} 0.75(1 - x^2)\, dx$$

$$= 0.75\left(x - \frac{x^3}{3}\right)\Big|_{0.5}^{0.5}$$

$= 0$

Note:

For a continuous variable, probability of a single point is zero.

$P(X = a) = 0$ for any real number a.

(f)

$P(X = 0.8) = 0$ (since X is continuous)

Note:

$f(x)$ is NOT probability. It gives density at $X = x$. In fact, $f(x)$ can be greater than one. Consider the pdf below (from Exercise 1).

$f(x) = 2x, \;\; 0 < x < 1.$

Here, $f(0.7) = 2 \times 0.7 = 1.4$ which is greater than one.

Cumulative distribution function
(or distribution function or cdf)

The cdf of a random variable X, denoted by $F(x)$, is defined as:

$F(x) = P(X \leq x)$

Example:

Consider the following pdf.

$$f(x) = \frac{1}{9}x^2, \;\; 0 < x < 3.$$

Determine (a) the cdf (b) $P(X \leq 1.5)$ (c) $P(X < 2.1)$ (d) $P(X < 3.7)$ (e) $P(X < -1.1)$ (f) $P(X > 2.0)$ (g) $P(1.5 < X < 2.0)$

Solution:

(a)

$$F(x) = \int_{-\infty}^{x} f(u)\, du$$

$$= \int_{0}^{x} \frac{1}{9}u^2\, du$$

$$= \frac{1}{27}x^3$$

Thus,

$$F(x) = \begin{cases} 0, & x \leq 0 \\ \frac{1}{27}x^3, & 0 < x < 3 \\ 1, & x \geq 3 \end{cases}$$

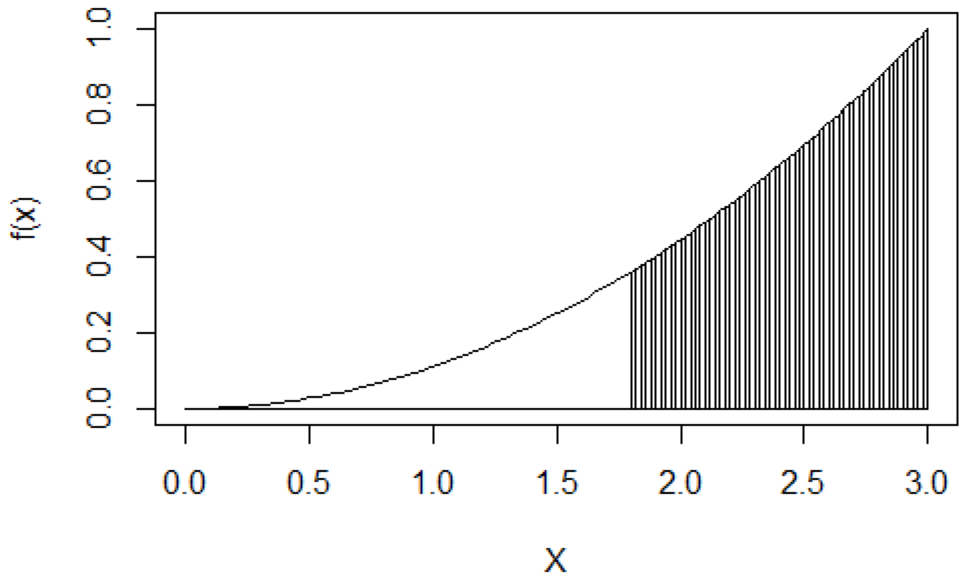
(b) $P(X \leq 1.5) = F(1.5) = \frac{1.5^3}{27} = 0.125$

(c) $P(X \leq 2.1) = F(2.1) = 0.343$

(d) $P(X < 3.7) = F(3.7) = 1$

(e) $P(X < -1.1) = F(-1.1) = 0$

(f) $P(X > 2.0) = 1 - P(X \leq 2.0) = 1 - F(2.0) = 1 - 0.296 = 0.704$



(g) $P(1.5 < X < 2.0) = F(2.0) - F(1.5) = 0.296 - 0.125 = 0.171$

Example:

Consider the following pdf.

$f(x) = 2x, \ \ 0 < x < 1.$

Determine (a) the cdf (b) $P(X \leq 0.5)$ (c) $P(X < 0.7)$ (d) $P(X < 1.9)$
(e) $P(X < -0.4)$ (f) $P(X > 0.6)$ (g) $P(0.5 < X < 0.7)$.

Solution:

Do it yourself.